

By Defn. 2, $\exists$ integers a, b where $b \neq 0$ such that $r = \frac{a}{b}$.
 Substituting the value of r in the equation $[r^3 + r + 1 = 0]$, we get:

$$\left(\frac{a}{b}\right)^3 + \left(\frac{a}{b}\right) + 1 = 0 \Rightarrow \frac{a^3}{b^3} + \frac{a}{b} + 1 = 0 \Rightarrow a^3 + ab^2 + b^3 = 0$$

Suppose, the fraction $r = \frac{a}{b}$ is in lowest terms, i.e. they have no common factors.

The L.H.S must have an even parity, since $RHS = 0$ (Even)

Case 1: a is even, b is even. By Defn 1, $\exists$ integers K_1 and K_2 such that $a = 2K_1$ and $b = 2K_2$. $\therefore a^3 + ab^2 + b^3 = (2K_1)^3 + 2K_1(2K_2)^2 + (2K_2)^3$
 $= 8K_1^3 + 8K_1K_2^2 + 8K_2^3$
 $= 2(4K_1^3 + 4K_1K_2^2 + 4K_2^3)$

By Defn 1, $a^3 + ab^2 + b^3$ has even parity which matches the parity of RHS.

Case 2: a is even, b is odd. By Defn 1, $\exists$ integers K_1 and K_2 such that $a = 2K_1$ and $b = 2K_2 + 1$. $\therefore a^3 + ab^2 + b^3 = (2K_1)^3 + 2K_1(2K_2 + 1)^2 + (2K_2 + 1)^3$
 $= 8K_1^3 + 2K_1(4K_2^2 + 4K_2 + 1) + (8K_2^3 + 12K_2^2 + 6K_2 + 1)$
 $= 2(4K_1^3 + 4K_1K_2^2 + 4K_1K_2 + K_1 + 4K_2^3 + 6K_2^2 + 3K_2) + 1$

$$= 2(4K_1^3 + 4K_1K_2^2 + 4K_1K_2 + K_1 + 4K_2^3 + 6K_2^2 + 3K_2) + 1$$

It's easy to see that $4K_1^3 + 4K_1K_2^2 + 4K_1K_2 + K_1 + 4K_2^3 + 6K_2^2 + 3K_2$ is an integer

By Defn 1, $a^3 + ab^2 + b^3$ has odd parity which contradicts the parity of RHS.

Case 3: a is odd, b is even. By Defn 1, $\exists$ integers K_1 and K_2 such that $a = 2K_1 + 1$ and $b = 2K_2$. $\therefore a^3 + ab^2 + b^3 = (2K_1 + 1)^3 + (2K_1 + 1)(2K_2)^2 + (2K_2)^3$
 $= (8K_1^3 + 12K_1^2 + 6K_1 + 1) + 4K_2^2(2K_1 + 1) + 8K_2^3$
 $= 2(4K_1^3 + 6K_1^2 + 3K_1 + 4K_1K_2^2 + 2K_2^2 + 4K_2^3) + 1$

It's easy to see that $4K_1^3 + 6K_1^2 + 3K_1 + 4K_1K_2^2 + 2K_2^2 + 4K_2^3$ is an integer.

By Defn 1, $a^3 + ab^2 + b^3$ has odd parity which contradicts the parity of RHS.